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Toward calibrated AI readiness in K–12 school-based practice: the roles of AI-related epistemic vigilance capacity and governance clarity

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The responsible use of generative artificial intelligence (AI) in K–12 schools requires professionals to evaluate AI-generated information while maintaining ethical boundaries and accountability for decisions. This two-phase study examined calibrated AI readiness, defined as self-reported readiness to verify AI outputs, recognize ethical boundaries, and retain responsibility for AI-assisted professional decisions. In Phase 1, 425 school-based professionals completed a survey assessing AI-related epistemic vigilance capacity, autonomous regulatory motivation, governance clarity, and calibrated AI readiness. Structural equation modeling indicated that epistemic vigilance capacity ($\beta = .675$, $p < .001$) and governance clarity ($\beta = .270$, $p < .001$) were positively associated with calibrated AI readiness, whereas autonomous regulatory motivation was not a significant predictor ($\beta = .076$, $p = .156$). An exploratory observed-score analysis found no significant interaction between epistemic vigilance capacity and autonomous regulatory motivation. In Phase 2, 25 school psychologists and school counselors participated in a one-month online pilot training program delivered through Moodle and Zoom. Self-reported calibrated AI readiness, measured using the same nine items as in Phase 1, increased descriptively from a pre-training mean of 4.20 to a post-training mean of 4.84 on a five-point scale. Scenario-based responses showed evidence of verification, privacy protection, contextual evaluation, and human accountability, although performance varied across specific indicators. Changes in self-reported readiness were not significantly associated with the continuous scenario-based reasoning index (Spearman's $\rho = -.02$, $p = .906$). Together, the findings suggest that perceived evaluative capacity and clear organizational guidance are important correlates of calibrated AI readiness, while also highlighting the value of combining self-report and applied reasoning assessments. Because Phase 1 was cross-sectional and Phase 2 lacked a comparison group, the results do not establish causal effects.

KEYWORDS
AI calibration, epistemic vigilance, generative artificial intelligence, governance clarity, human-AI collaboration, responsible AI use

1 Introduction

Large language models can generate coherent and contextually appropriate text through statistical prediction, but they may also produce inaccurate information, fabricate references, reproduce embedded biases, or generate recommendations that are inappropriate for a particular context (Ferrara, 2023; Nafar et al., 2025). These limitations are especially consequential in K–12 educational practice, where generative artificial intelligence (AI) is increasingly used to support documentation, communication, instructional planning, intervention development, and other language-intensive professional tasks (Farmer et al., 2025; Lockwood et al., 2026). Because such activities may contribute to educational decisions involving student rights, confidentiality, and access to services, school professionals must retain responsibility for the accuracy, legality, and ethical appropriateness of AI-assisted work.

The increasing accessibility of generative AI does not diminish professional responsibility. Instead, it heightens the cognitive, ethical, and organizational demands placed on professional judgment. Responsible engagement requires professionals not only to understand what AI systems can do, but also to determine when AI-generated information is sufficiently credible, appropriate, and ethically permissible to inform practice. This concern shifts attention away from AI adoption or generalized trust and toward calibrated reliance. Calibrated reliance is a kind of capacity to determine when AI-generated recommendations should be accepted, independently verified, modified, or rejected (Okamura and Yamada, 2020; Schemmer et al., 2023; Schoeffer et al., 2023).

Research on human–AI interaction has shown that greater trust in AI does not necessarily produce more responsible use. For those users who have greater trust in AI, they may tend to be more likely to accept AI-generated suggestions without critical verification (Dogru and Krämer, 2025; Küper and Krämer, 2025). However, most existing research has been conducted in controlled laboratory settings using tasks that typically have objectively correct answers. In contrast, within actual school practices, professional decision-making often requires interpretation based on specific contexts while simultaneously accounting for ethical sensitivities and legal responsibilities. Consequently, there remains a lack of sufficient research on how to understand and define Calibrated AI readiness in such real-world practice settings.

These individual- and organizational-level conditions are usually examined within distinct research traditions. Educational research primarily focuses on the adoption, implementation, attitudes toward, or literacy regarding AI (Kim, 2026; Ma et al., 2025). Research on human-AI interaction often investigates calibrated reliance through controlled experimental tasks (Dogru and Krämer, 2025; Küper and Krämer, 2025; Schemmer et al., 2023). Studies on responsible AI emphasize aspects such as governance, transparency, privacy, fairness, human oversight, and accountability (Farmer et al., 2025; Papagiannidis et al., 2022). However, few studies integrate these perspectives to explain how professionals in school settings develop the capacity to use generative AI responsibly while upholding evidence-based decision-making, ethical authority, and human accountability.

To fill in this research gap, this study provides an epistemic-governance framework. At the individual level, this framework emphasizes AI-related epistemic vigilance capacity which means the school professionals can verify credibility, plausibility, evidentiary support, and contextual appropriateness of AI-generated information. At the same time, it includes the autonomous regulatory motivation that refers to the professionals' willingness to engage in continuous learning and self-directed improvement when utilizing emerging technologies (Deci and Ryan, 2000). At the organizational level, this framework focuses on governance clarity that is if professionals perceive their organizations' guidance on AI use policy, expectation of using AI, ethic boundary, and accountability procedures are clear and adequate.

Therefore, this study focused on the key conditions that support K-12 school professionals who use AI in ethical and calibrated ways. Based on the proposed epistemic-governance framework, this study in phase 1 examined the relationships between AI-related epistemic vigilance capacity, autonomous regulatory motivation, governance clarity, and calibrated AI readiness. In phase 2, this study piloted an online training about responsible AI use with a group of school professionals. The training evaluated post-training descriptive changes and applied reasoning performance through self-reports and evidence from situational tasks.

2 Literature review

2.1 Generative AI in school-based professional practice

In K-12 schools, school professionals used AI to draft reports, generate students information, develop intervention ideas, support lesson plans, and facilitate the communications with families (Farmer et al., 2025; Lockwood et al., 2026). These AI applications are helpful for improving school professionals' efficiency, reducing administrative burden, and particularly helpful for positions that have a high demands for significant documentation and communication. Nevertheless, the generative AIs are not based on deep understanding or independent evaluation to generate outputs, instead, it depends on probability. It may lead to hallucinations, fake citations, embedded bias, or not adequately considering individual students' educational, cultural, developmental, and legal contexts (Ferrara, 2023; Nafar et al., 2025). These limitations are particularly important in school settings, as written documentation and professional suggestions might influence students' eligibility determinations, intervention planning, family communication, and other significant decisions.

Legal and ethical obligations within school settings further amplify these risks. The documents or suggestions produced by AI might not be compliant with the confidentiality requirements made by the Individuals with Disabilities Education Act (IDEA) or Family Educational Rights and Privacy Act (FERPA) (Farmer et al., 2025). Even though AI can assist in handling information processing and writing documents, the responsibilities of professional decisions cannot be put under the automated systems (Farmer et al., 2025; Lockwood and Brown, 2026). It is

still required that professionals should examine the information generated by AI, evaluate the accuracy, appropriateness, compliance with law requirement, what is more important, tailored to the specific needs of students or their families. The emerging evidence shows that the professionals' adoption of using AI in work has outpaced the development of domain governance guidance. In K-12 schools, there are obvious differences in organizational policy, AI-use disclosure requirements, informed consent, professional development in AI, governance procedures among school professionals (Lockwood et al., 2026). Therefore, the key is not whether the professionals can use AI or not, but if they are able to manage their reliance on AI within the clear ethical and organizational boundaries.

2.2 Human–AI calibration and responsible AI judgment

Research on human-AI interaction has introduced the concept of calibrated reliance, wherein users adjust their level of reliance based on the actual quality and contextual appropriateness of AI-generated information. Individuals should evaluate which suggestions to adopt, which require further verification, and which should be rejected (Okamura and Yamada, 2020; Schemmer et al., 2023). Thus, appropriate reliance reflects a modulation of human judgment rather than unconditional trust in AI.

Experimental studies show that greater trust in AI might increase the possibilities of users' adoption of AI suggestions, but without or short of critical evaluation capabilities, it might also increase user's likelihood to accept the inaccurate outputs (Dogru and Krämer, 2025; Küper and Krämer, 2025; Schemmer et al., 2023). Besides, we cannot decide if the reliance on AI is appropriate or not based on whether users' final decision is correct or not. Even though two users might make the same conclusion of accepting or rejecting AI outputs, the judging process might be significantly different. It ranges from making sound, independent 106 judgments to over-relying on flawed AI outputs or rejecting valuable AI assistance (Schoeffer et al., 2023). Consequently, assessing responsible AI usage should focus on how individuals calibrate their reliance on AI, 108 rather than merely on the correctness of their final answers.

Existing evidence largely stems from laboratory tasks, such as chess, hotel review classification, and factchecking, that typically feature relatively clear standards of correctness (Dogru and Krämer, 2025; Schemmer et al., 2023). In contrast, professional decision-making within educational settings often involves incomplete information, multiple plausible interpretations, ethical trade-offs, contextual knowledge, and legal liabilities. An AI recommendation, even if technically sound, may not be appropriate for a specific student, family, school, or decision-making context.

Calibration also relates to if AI users position AI as the source of knowledge and authority. Rind et al. (2026) found that on the one hand, AI users unquestionably accept the AI-generated outputs without verifications and critical examination and therefore develop over-reliance on AI for the purpose of efficiency. On the other hand, some AI users maintain strategic skepticism of AI with critical thinking and evaluation. Users with high level of calibration engagement demonstrate capability in adjusting reliance on AI and verifying AI outputs depending on the contexts of tasks. These users also treat the AI outputs as provisional rather than authoritative. Although this study focuses on college students not K-12 school professionals, it is still helpful to clarify the epistemic processes underlying critical engagement with AI. Within the proposed conceptual model in this study, this skepticism, verification, and evaluation are closely related to the AI-related epistemic vigilance capacity.

Based on this distinction, this study defines calibrated AI readiness as professionals' capability to adjust their reliance on AI generated outputs. Specifically, this capability includes systematically verifying the AI outputs, discerning ethical and privacy boundaries, critically evaluating the uncertainties, and preserving the accountability ownership. calibrated AI readiness does not mean using AI in every contexts, but emphasizing that the professional can make right decisions about when, where and how to use AI appropriately as assistance tool, and avoid using AI to replace professional judgment. Rind (2026) conceptualizes teachers as cognitive orchestrators. Teachers need to keep a balance of the algorithmic assistances with diagnostic reasoning, contextual interpretation, and ethical judgment. From this perspective, AI may reduce routine or extraneous demands while also displacing the reflective and diagnostic work that sustains professional expertise. Consistent with this argument, teachers in Rind and Qureshi's (2026) qualitative study described AI as providing cognitive relief but also shifting instructional sequencing, feedback interpretation, and evaluative judgment toward algorithmic systems. Professional agencies were not eliminated but became conditional on teachers' capacity and institutional freedom to question, adapt, or override AI recommendations. Calibrated AI readiness therefore involves not only detecting inaccurate outputs but also retaining interpretive control over whether and how AI informs professional action.

2.3 Individual determinants of calibrated AI readiness

Calibrated AI readiness is related to, but conceptually distinct from, several constructs concerning AI knowledge, confidence, evaluation, reliance, and professional decision-making. Table 1 clarifies the boundaries of the proposed construct and its relationship with neighboring concepts.

These distinctions position AI-related epistemic vigilance as a specific evaluative capacity that may support calibrated AI readiness. In contrast, calibrated AI readiness represents a broader professional orientation that combines verification, ethical boundary recognition, and accountability for AI-assisted decisions. The conceptual overlap between epistemic vigilance and verification orientation is therefore expected, but their empirical distinctiveness requires direct examination.

2.3.1 AI-related epistemic vigilance capacity

This study posits that epistemic vigilance is a crucial individual capability enabling school professionals to develop a

TABLE 1 Conceptual boundaries of calibrated AI readiness and related constructs.

Construct	Conceptual definition	Primary dimensions or focus	Theoretical basis	Relationship to and distinction from calibrated AI readiness
Calibrated AI readiness	A school professional's readiness to determine when and how AI-generated information should be verified, modified, used, or rejected while maintaining ethical boundaries and human accountability.	Verification orientation; ethical boundary clarity; accountability ownership.	Calibrated reliance, epistemic vigilance, ethical competence, and accountability ownership (De Schrijver and Maesschalck, 2013; Okamura and Yamada, 2020; Sperber et al., 2010).	The focal construct in this study. It integrates evaluative, ethical, and accountability orientations rather than focusing only on AI knowledge, confidence, trust, or task performance. In this study, it is operationalized as perceived readiness rather than demonstrated competence.
AI-related epistemic vigilance capacity	The perceived capacity to evaluate the credibility, plausibility, evidentiary support, uncertainty, and contextual appropriateness of AI-generated information before accepting it (Eyal, 2025; Sperber et al., 2010).	Verification of evidence; recognition of inaccuracies and uncertainty; evaluation of sources; regulation of reliance.	Epistemic vigilance and critical evaluation of communicated information (Sperber et al., 2010).	It is conceptualized as a narrower evaluative capacity that may support calibrated AI readiness. It overlaps with verification orientation because both involve checking AI outputs; however, calibrated AI readiness additionally includes ethical boundaries and accountability. This theoretical distinction requires empirical discriminant-validity evidence.
AI literacy	Knowledge and skills needed to understand, use, evaluate, and critically engage with AI systems and their social and ethical implications (Eyal, 2025; Ma et al., 2025; UNESCO, 2024).	Technical understanding; practical use; critical evaluation; ethical and social awareness.	Digital and AI literacy and competency frameworks.	AI literacy is broader in its coverage of knowledge and skills and may apply across educational and everyday contexts. Calibrated AI readiness focuses more specifically on professional decisions about reliance, verification, ethical boundaries, and accountability in AI-assisted practice.
AI self-efficacy	An individual's belief in their capability to learn about or successfully perform tasks using AI tools (Bergdahl and Sjöberg, 2025; Du et al., 2026).	Perceived confidence; capability beliefs; willingness to perform AI-related tasks.	Self-efficacy and social-cognitive perspectives on technology use.	Self-efficacy concerns confidence in one's ability to use AI. High confidence does not necessarily indicate that a professional can identify inaccurate outputs, recognize ethical risks, or calibrate reliance appropriately.
Responsible AI competence	A broad set of knowledge, skills, and dispositions supporting ethical, transparent, fair, privacy-protective, and human-centered AI use (Adams et al., 2023; Ma et al., 2025; UNESCO, 2024).	Ethics; fairness; transparency; privacy; human oversight; responsible application.	Responsible AI and AI ethics competency frameworks.	Responsible AI competence is a broad umbrella covering technical, social, and ethical capabilities. Calibrated AI readiness is a more focused proposed construct concerning professional readiness to evaluate AI outputs, recognize boundaries, and retain responsibility for AI-assisted decisions.
Trust calibration and appropriate reliance	The alignment of a user's trust and reliance with the actual quality, reliability, and contextual appropriateness of AI assistance (Okamura and Yamada, 2020;	Acceptance, verification, modification, or rejection of AI advice according to its quality and the decision context.	Human–AI interaction, automation trust, and reliance calibration.	Trust calibration is commonly examined through reliance behavior within specific decision tasks. Calibrated AI readiness concerns professionals' broader readiness to

(*Continued*)

TABLE 1 Continued

Construct	Conceptual definition	Primary dimensions or focus	Theoretical basis	Relationship to and distinction from calibrated AI readiness
	Schemmer et al. 2023; Schoeffer et al., 2023).			make such judgments across uncertain school-based situations while also considering ethics, privacy, and accountability.
Professional judgment	The context-sensitive application of expertise, evidence, ethical standards, and professional responsibility when making decisions under uncertainty (De Schrijver and Maesschalck, 2013; Farmer et al., 2025; Karakuş et al., 2025).	Interpretation of evidence; contextual reasoning; ethical decision-making; responsibility for final decisions.	Professional expertise, ethical competence, and evidence-informed decision-making.	Professional judgment is a broader human decision-making process that applies with or without AI. Calibrated AI readiness concerns the preservation and application of that judgment when AI-generated information enters professional practice.

Calibrated AI readiness is proposed as a multidimensional construct in the present study. The distinctions summarized here are theoretical and do not imply complete empirical independence among the constructs.

calibrated readiness for AI. Rooted in communication theory, epistemic vigilance refers to the cognitive mechanism individuals employ to assess the credibility of information before accepting it as true (Sperber et al., 2010). It entails neither the unquestioning acceptance of information nor generalized skepticism; rather, it involves evaluating the credibility of the source alongside the plausibility, consistency, evidentiary support, and relevance of the content.

In the context of generative AI, epistemic vigilance requires school professionals to actively scrutinize AI-generated information rather than mistaking fluent or confident expression for evidence of accuracy. This process involves verifying supporting evidence, identifying uncertainties, comparing AI outputs against professional knowledge, assessing the credibility of cited sources, and determining the applicability of recommendations to specific educational contexts (Eyal, 2025). Furthermore, professionals must recognize that AI systems can generate persuasive responses even when the underlying information is incomplete or inaccurate (Salvi et al., 2025).

Recent research on human-AI interaction also supports the importance of evaluative capabilities. Appropriate reliance appears to depend more on a user's ability to evaluate AI-generated information and their own judgment, rather than merely on confidence, technological affinity, or general trust (Dogru and Krämer, 2025; Küper and Krämer, 2025). Individuals who can more accurately identify the respective strengths and limitations of AI and human judgment are better equipped to determine when AI assistance should influence professional decision-making. Consequently, epistemic vigilance may constitute the cognitive foundation for the responsible use of AI by school professionals.

Therefore, in this study, AI-related epistemic vigilance defines school professionals' ability to critically evaluate the AI outputs. It includes verifying the supporting evidence of AI claims, recognizing the uncertainness in outputs, and evaluating the contextual appropriateness of AI-assisted suggestions. Professionals with higher degree of epistemic vigilance will be more likely to discern unreliable AI outputs and accordingly adjust their acceptance of AI outputs.

2.3.2 Autonomous regulatory motivation

Motivation can also influence how professional approach AI-related learning and making decisions with assistance of AI. Self-determination theory (Deci and Ryan, 2000) distinguishes between autonomous and controlled motivation. Within this theory, autonomous regulatory motivation refers to individual motivation to engage elicited by meaningfulness, value, or alignment with professional goals. Contrary to this, controlled motivation is primarily from external pressure, organizational requirement, or the need of avoiding negative consequences.

The autonomous regulatory motivation is particularly important when professional is dealing with new technologies that are short of established policies, standardized practices, and sufficient training opportunities. If professionals think it is important to responsibly learn and use AI for professional development, they are more likely to proactively accrue related information, evaluate the potential risks, participate in professional training, and continuously improve self-practices (Schumann et al., 2025). The existing studies about educational technology also show that autonomous regulatory motivation is positively associated with technology acceptance and continuous professional learning (Kim, 2026).

However, autonomous regulatory motivation alone is not sufficient for professionals to adjust their reliance on AI outputs. Even if professionals have high degree of motivation and actively use AI, it is possible that they still accept inaccurate outputs, overlook the risk of privacy exposure, or rely on AI outputs in unsuitable contexts. Some human-AI interaction studies also point out that positive attitude, engagement willingness, sense of trust do not guarantee appropriate reliance on AI (Dogru and Krämer, 2025; Küper and Krämer, 2025). Therefore, even though autonomous regulatory motivation plays a positive role in contributing to professionals' responsible use of AI and ethical engagement (Jansen in de Wal et al., 2014), its role must be further examined along with the epistemic vigilance of professionals.

This study further examines whether autonomous regulatory motivation remains associated with calibrated AI readiness after controlling for AI-related epistemic vigilance. This distinction is crucial because professionals differ not only in their willingness to engage in responsible AI learning but also in their ability to effectively evaluate AI-generated information.

2.4 Organizational context and governance clarity

Responsibly using AI is a sociotechnical challenge, which relies on individual capacity and organizational conditions. Professionals' practices are shaped by institutional policies, professional standards, supervisory expectations, privacy procedures, training opportunities, accountability structures, and the organizational climate regarding technology use (National Institute of Standards and Technology [NIST], 2023; UNESCO, 2024).

The existing frameworks for responsible AI use generally focus on the importance of transparency, fairness, privacy protection, human oversight, accountability, institutional governance, and continuous professional learning (Farmer et al., 2025; Papagiannidis et al., 2022; Wickramasinghe et al., 2026). These principles are particularly important in education. Because teachers' use of AI can influence students' educational opportunities and rights to confidentiality which are protected by legal and professional standards. Therefore, clear governance can provide decision-making frameworks for professionals which enable them to make informed judgments while adhering to ethical and legal requirements.

Governance clarity refers to the extent to which an organization's expectations regarding AI use are easily understood, accessible, and specific enough to guide professional practice (National Institute of Standards and Technology [NIST], 2023). This includes rules governing data use, disclosure, verification, accountability, and acceptable AI practices. Governance clarity is particularly crucial in educational settings, where professionals must navigate a complex array of legal, ethical, and organizational requirements. Recent studies indicate significant disparities among school professionals regarding AI-related policies, disclosure practices, informed consent procedures, and formal training (Lockwood et al., 2026). When governance is unclear, professionals may rely on individual judgment, leading to inconsistent practices. Thus, governance clarity may contribute to differences in AI readiness.

2.5 Research gap and present study

The existing studies show that school professionals' meaningful and ethical use of AI in work related practices are not only bound to their technology access, frequency of use, knowledge of using AI, or favorable attitudes. Calibrated AI readiness is likely to depend on the interaction of individual evaluative capabilities, autonomous regulatory motivation, and organizational guidance. But few studies have integrated these perspectives into a unified framework explaining how school-based professionals develop the capacity to engage responsibly with generative AI in authentic educational practice.

To address this gap, the present study integrates these dimensions within a conceptual model of calibrated AI readiness and examines their relevance across two complementary phases. Specifically, Phase 1 examines whether AI-related epistemic vigilance capacity, autonomous regulatory motivation, and governance clarity predict calibrated AI readiness among school psychologists and related school-based professionals. Phase 2 extends this framework by piloting a structured responsible AI training program and examining descriptive changes in self-reported readiness together with evidence of applied reasoning from scenario-based tasks.

2.6 Conceptual model of calibrated AI readiness

The proposed conceptual model was developed by integrating four complementary theoretical and empirical perspectives. First, research on calibrated reliance in human–AI collaboration informed the conceptualization of calibrated AI readiness as the capacity to evaluate when AI-generated recommendations should be accepted, verified, modified, or rejected (Okamura and Yamada, 2020; Schemmer et al., 2023; Schoeffer et al., 2023). Second, epistemic vigilance theory provided the foundation for examining professionals' perceived capacity to critically evaluate the credibility, evidentiary support, and contextual appropriateness of AI-generated information (Sperber et al., 2010). Third, Self-Determination Theory informed the inclusion of autonomous regulatory motivation as a potential contributor to responsible AI engagement (Deci and Ryan, 2000, 2008). Finally, responsible AI governance frameworks informed the inclusion of governance clarity as an organizational condition that may support ethical decision-making, human oversight, and accountability (National Institute of Standards and Technology, 2023; UNESCO, 2024). By integrating these perspectives, the present study proposes an epistemic–governance model that considers both individual and organizational conditions associated with self-reported calibrated AI readiness in K–12 school-based practice.

Building on the preceding literature, this study proposes a conceptual model of calibrated AI readiness. The framework positions calibrated AI readiness as the focal professional outcome and integrates individual and organizational conditions expected to support responsible engagement with generative AI. In other words, the model examines whether professionals report greater readiness to use AI responsibly when they are better able to critically evaluate AI-generated information, are personally motivated to engage in responsible AI learning, and receive clearer organizational guidance.

At the individual level, AI-related epistemic vigilance capacity was expected to be positively associated with calibrated AI readiness because professionals who more critically evaluate AI-generated information should be better able to regulate their reliance on it. Therefore, autonomous regulatory motivation was hypothesized to have a positive association with calibrated AI readiness in this study. In addition to its hypothesized direct association, autonomous regulatory motivation was examined as

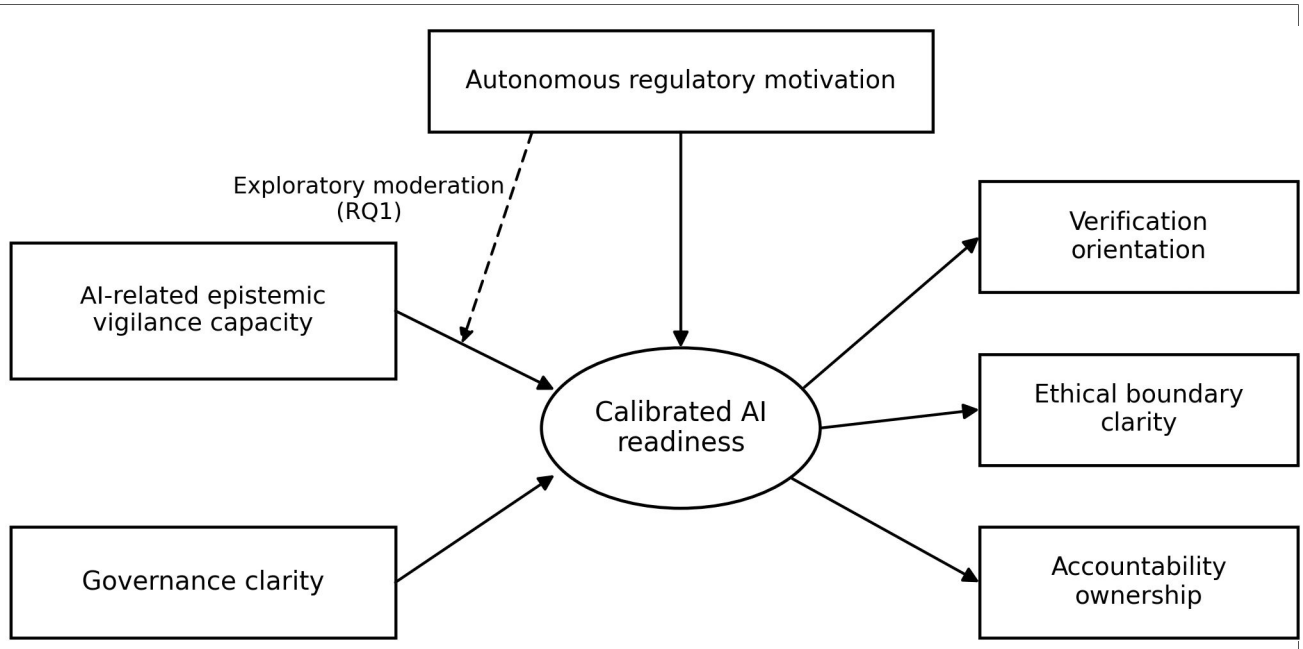


FIGURE 1
Proposed conceptual model of calibrated AI readiness.

a potential moderator of the relationship between epistemic vigilance capacity and calibrated AI readiness. This moderating relationship was treated as exploratory because the existing literature does not provide sufficient evidence to specify a directional interaction. At the organizational level, governance clarity was expected to be positively associated with calibrated AI readiness by providing clearer expectations, boundaries, and accountability structures for professional AI use.

Figure 1 presents the conceptual model tested in Phase 1. Phase 2 extends the framework by examining whether structured responsible AI training is associated with changes in calibrated AI readiness and scenario-based reasoning.

The proposed model tested the following hypotheses and research questions.

2.6.1 Hypotheses

H1: AI-related epistemic vigilance capacity will be positively associated with calibrated AI readiness ($\beta 1 > 0$).

H2: Autonomous regulatory motivation will be positively associated with calibrated AI readiness ($\beta 2 > 0$).

H3: Governance clarity will be positively associated with calibrated AI readiness, controlling for epistemic vigilance capacity and autonomous regulatory motivation ($\beta 3 > 0$).

2.6.2 Research questions

RQ1. Does autonomous regulatory motivation moderate the association between AI-related epistemic vigilance capacity and calibrated AI readiness?

RQ2. What descriptive changes in training-aligned self-reports and what patterns of scenario-based reasoning are observed during the pilot training?

3 Materials and methods

3.1 Research design

This study examined individual and contextual conditions associated with ethically grounded and professionally responsible engagement with generative AI tools in school-based practices. A two-phase, multimethod research design was developed to examine both the correlates of calibrated AI readiness and the preliminary application of the proposed conceptual framework in professional learning. The design was informed by prior research on calibrated human–AI reliance, epistemic vigilance, and responsible AI governance (Okamura and Yamada, 2020; Schemmer et al., 2023; Sperber et al., 2010; UNESCO, 2024). However, the specific integration of a cross-sectional structural modeling study and a pilot professional learning evaluation was developed to address the research objectives of the present study.

Phase 1 used a cross-sectional survey design to test a structural equation model in which AI-related epistemic vigilance capacity and governance clarity were examined as predictors of calibrated AI readiness, and autonomous regulatory motivation was tested as a direct predictor. Its potential moderating role was examined separately through an exploratory observed-score analysis. Phase 2 used a single-group pretest–posttest pilot training design to examine feasibility, descriptive change, and the relationship between changes in self-reported calibrated AI readiness and scenario-based responsible AI reasoning.

These two phases served complementary purposes. Phase 1 examined associations among the theoretically proposed constructs in a larger sample of school-based professionals, whereas Phase 2 explored the practical application of the framework through structured training and scenario-based assessments. The inclusion of both self-report and applied reasoning measures in Phase 2 allowed the study to examine

whether perceived changes in readiness corresponded with demonstrated reasoning in AI-related professional situations. Because Phase 1 was cross-sectional and Phase 2 did not include a comparison group, neither phase was intended to establish causal relationships or training effectiveness.

3.2 Phase 1: SEM study

3.2.1 Participants and recruitment

Participants were practicing school psychologists and related school-based professionals, including supervisors, school counselors, teachers and special education professionals working in K-12 settings. Eligibility criteria required participants to be at least 18 years old and actively practicing in a school-based professional role. Participants were recruited using publicly available online school district directories. Invitation emails were sent directly to practicing school-based professionals identified through district websites.

Participants completed a 10- to 15-minute online Qualtrics survey assessing calibrated AI readiness, AI-related epistemic vigilance capacity, autonomous regulatory motivation, governance clarity, baseline AI use, and professional context variables. Participation was voluntary. Respondents could optionally provide contact information to be invited to Phase 2. To protect confidentiality, contact information was stored separately from survey responses. Survey data were collected between April and May 2026. The final Phase 1 analytic sample included 425 respondents. Item-level data were complete for all measures except autonomous regulatory motivation. Of the 425 respondents, 420 completed all three autonomous regulatory motivation items. The item-level SEM retained all 425 respondents using full-information maximum likelihood estimation. Participants with complete data were entered into a drawing for one of ten e-gift cards as a token of appreciation for their participation in the study. The details of participants in this study can be found in Table 2.

3.2.2 Measures

No single existing measure captured the combination of verification, ethical and privacy boundaries, and professional accountability examined in this study within K–12 professional practice. Therefore, I developed all the scales used in this study (see Supplementary Material A for the scales). After reading the studies about human-AI collaboration, epistemic vigilance, responsible AI, Self-Determination Theory, and professional ethics in educational practice, I developed the items of each scale to reflect the authentic school-based decision-making contexts. The draft items were revised and modified iteratively after consulting and discussion with one educational methodologist and two K-12 schoolteachers to improve conceptual clarity, relevance, and content coverage. This process constituted a preliminary qualitative review of item content rather than a formal content-validation study; no quantitative content-validity index was calculated. All the items were rated on a five-point Likert scale ranging from 1 (strongly disagree) to 5 (strongly agree). The higher score shows higher level of endorsement in this construct. The composite score for each variable used for data analysis was computed by averaging the item responses.

3.2.2.1 Calibrated AI readiness

Calibrated AI readiness is the key and primary outcome variable in this study. It was assessed through a nine-item study-specific scale comprising three dimensions. They included verification orientation (3 items), ethical boundary clarity (3 items), and accountability ownership (3 items).

The dimension of verification orientation subscale was used to assess school professional's tendency to verify the AI-generated outputs and detect the uncertainty before accepting AI's outputs. Ethical boundary clarity was used to assess school professionals' understanding about how to appropriately and ethically use AI in their practices, protect privacy and confidentiality. Accountability ownership measured participants' understanding about taking responsibility for AI-assisted professional decision making, and their capability to explain and justify their AI use aligning with the professional standards. I computed the average score of these nine items to show the overall calibrated AI readiness score.

3.2.2.2 AI-related epistemic vigilance capacity

AI-related epistemic vigilance capacity was measured using a three-item study-specific scale assessing participants' tendency to critically evaluate AI-generated information before incorporating it into professional practice. Three items assessed verification behaviors, recognition of potential inaccuracies, identification of warning signs in AI-generated responses, and appropriate regulation of reliance on AI across professional contexts. Higher scores indicated stronger perceived capacity to critically evaluate AI-generated information.

3.2.2.3 Autonomous regulatory motivation

Autonomous regulatory motivation was measured using three study-specific measures items conceptually adapted from Self-Determination Theory (Deci and Ryan, 2000, 2008). The three items assessed the extent to which participants viewed responsible AI use as personally meaningful, supportive of professional growth, and beneficial for serving students and families. The higher score represented that participants had higher motivation to engage in responsible AI use.

3.2.2.4 Governance clarity

The scale of governance clarity has three items, which were used to assess school professionals' perceptions of organizational guidance on AI use. These items assessed participants' clarity about institutional expectations, understanding of how and to what extent to use AI in works, and their knowledge about the requirement of disclosure of AI usage in AI-assisted schoolworks. Higher scores indicated greater perceived clarity of organizational governance for AI use.

3.2.3 Reliability and measurement quality

Internal consistency was evaluated using Cronbach's alpha and model-based composite reliability. Convergent validity was examined using standardized factor loadings and average

TABLE 2 Participant background and AI-use characteristics by study phase.

Study phase	Characteristic	Category	*n*	%
Phase 1 ($N = 425$)	Years of professional experience	0–2 years	177	41.6
		3–5 years	119	28.0
		6–10 years	54	12.7
		11–20 years	40	9.4
		More than 20 years	35	8.2
	Frequency of generative AI use for work	Never	40	9.4
		Less than monthly	47	11.1
		Monthly	40	9.4
		Weekly	83	19.5
		Multiple times per week	97	22.8
		Daily or almost daily	118	27.8
	Workplace AI-policy status	Clear written policy	143	33.6
		Informal guidance only	150	35.3
		No guidance	58	13.6
		Unsure	63	14.8
		Unresolved coded response	10	2.4
		Missing	1	0.2
Phase 2 pilot ($N = 25$)	Professional role	School psychologist	15	60.0
		School counselor	10	40.0
	Years in current or similar role	0–2 years	11	44.0
		3–5 years	7	28.0
		6–10 years	2	8.0
		11–15 years	2	8.0
		16 years or more	3	12.0
	Current AI-use frequency	Rarely	5	20.0
		Sometimes	10	40.0
		Often	8	32.0
		Always	2	8.0
	AI familiarity	Slightly familiar	5	20.0
		Moderately familiar	7	28.0
		Very familiar	13	52.0
	Prior AI training	No prior training	6	24.0
		Brief exposure, such as a workshop or webinar	14	56.0
		Formal training or course	5	20.0
	Common AI-use areas	Lesson planning or instructional support	16	64.0
		Communication, such as emails or reports	14	56.0
		Student support, such as interventions or accommodations	13	52.0
		Administrative tasks	8	32.0

The Phase 2 common AI-use areas were assessed using a select-all-that-apply item; therefore, percentages do not sum to 100%.

variance extracted (AVE). Discriminant validity was evaluated using latent-factor correlations, the Fornell–Larcker criterion, and the heterotrait–monotrait ratio of correlations (HTMT).

Construct distinctiveness was further evaluated by comparing four measurement models: the proposed second-order model, a correlated six-factor model, a model combining epistemic vigilance and verification orientation, and a model representing the nine calibrated AI readiness items as a single factor. Model comparisons were based on robust fit indices and information criteria. These analyses provided evidence concerning reliability, internal structure, and construct distinctiveness, but they were not treated as substitutes for formal content-validity or response-process evidence.

3.2.4 Structural model specification

Structural equation modeling was conducted in R using the lavaan package, Version 0.6-21. Models were estimated using robust maximum likelihood estimation (MLR), with full-information maximum likelihood used to address missing data. MLR was selected because the study used Likert-type indicators and because it provides standard errors and fit indices that are less sensitive to violations of multivariate normality (Li, 2016; Rhemtulla et al., 2012). Because the measures used five-point Likert indicators, the primary second-order structural model was also re-estimated using weighted least squares mean and variance adjusted estimation (WLSMV) as a sensitivity analysis. All 18 item-level indicators were treated as ordered categorical variables. The sensitivity model used theta parameterization and pairwise-present data for the limited missing responses.

An item-level second-order SEM was estimated to represent the multidimensional structure of calibrated AI readiness. Verification orientation, ethical boundary clarity, and accountability ownership were specified as first-order latent factors, each measured by three items. These factors loaded onto a second-order calibrated AI readiness factor. AI-related epistemic vigilance capacity, autonomous regulatory motivation, and governance clarity were modeled as separate latent predictors, each measured by three items. This model was used as the primary structural model because it represented the proposed theoretical structure and accounted for measurement error.

As a supplementary exploratory analysis, moderation was examined using observed composite scores. The composite scores for AI-related epistemic vigilance capacity and autonomous regulatory motivation were mean-centered and multiplied to create an interaction term. The interaction term was entered with the corresponding main effects and governance clarity as predictors of calibrated AI readiness. Because the interaction was constructed from observed scores, this exploratory analysis did not explicitly account for measurement error and was not treated as a confirmatory test of moderation.

The moderated model was specified as follows:

$$\text{AI_READY} = \beta 0 + \beta 1(\text{EVC}) + \beta 2(\text{AUTO}) + \beta 3(\text{GOV}) + \beta 4(\text{EVC} \times \text{AUTO}) + \varepsilon$$

In this model, AI_READY represents calibrated AI readiness, EVC represents AI-related epistemic vigilance capacity, AUTO represents autonomous regulatory motivation, GOV represents governance clarity, and EVC × AUTO represents the interaction between epistemic vigilance capacity and autonomous regulatory motivation. The exploratory moderation analysis addressed RQ1 by examining the interaction coefficient, β4. A nonsignificant interaction was interpreted as an absence of evidence for moderation in the present analysis rather than definitive evidence that moderation was absent.

Model fit was evaluated using multiple indices, including the chi-square test of model fit, Comparative Fit Index (CFI), Tucker–Lewis Index (TLI), Root Mean Square Error of Approximation (RMSEA), and Standardized Root Mean Square Residual (SRMR). Nested models were compared using a scaled chi-square difference test appropriate for models estimated with MLR.

3.3 Phase 2: single-group pilot training evaluation

3.3.1 Design

Phase 2 used a single-group pretest and posttest training design to study if the calibrated AI readiness framework could be applied in authentic school works. This phase aimed to examine feasibility, descriptive within-person change, and convergence between self-reported calibrated AI readiness and scenario-based reasoning. Because the phase 2 design did not include a control group, Phase 2 did not aim to provide causal evidence of training effectiveness.

3.3.2 Participants

There were 25 school-based professionals who completed the pre-training survey, four weekly scenario-based training tasks, and the post-training survey. Participants were primarily school psychologists and school counselors with different levels of professional experience, AI familiarity, current AI use, and prior AI training. The details of participant characteristics can be found in Table 1. Most participants had some prior exposure to AI tools, with 76.0% reporting either brief exposure through a workshop or webinar or formal training/coursework. The most reported AI-use areas were lesson planning or instructional support, communication, and student support.

3.3.3 Training context and procedure

Participants completed a 1-month calibration-oriented responsible AI training program focused on AI use in educational and school-based professional practice. The program included four weekly sessions, with each session designed to take approximately 1 h. Training materials were hosted on Moodle, sessions were delivered through Zoom, and all surveys and embedded scenario tasks were collected through Google Forms. No Zoom recordings were collected because some participants opted out of recording. The four weekly sessions were organized around the three core dimensions of calibrated AI readiness. The details of the each weekly session can be found in Supplementary Material B.

The training sequence included six data-collection points: a pre-training survey, four weekly scenario-based training tasks, and a post-training survey. The pre-training survey assessed baseline AI use, AI familiarity, prior AI training, and calibrated AI readiness. The post-training survey repeated the core calibrated AI readiness items and included reflection items about perceived usefulness, awareness of AI-related risks, ethical concerns, verification, accountability, and safe-use strategies.

3.3.4 Measures

Self-reported calibrated AI readiness was assessed using the same nine items administered in Phase 1. The items represented the same three domains specified in the Phase 1 measurement model: verification orientation (three items), ethical boundary clarity (three items), and accountability ownership (three items). Items were coded from 1 = disagree to 5 = agree, with higher scores indicating greater self-reported calibrated AI readiness. Domain scores were calculated by averaging the three corresponding items, and an overall calibrated AI readiness composite was calculated by averaging all nine items. Because Phase 2 was a small pilot with 25 participants, a separate measurement model or longitudinal measurement-invariance analysis was not estimated.

3.3.5 Scenario-based reasoning

Scenario-based reasoning was assessed through open-ended responses from the four embedded training tasks. Responses were coded for evidence of applied reasoning in six areas: verification, contextualization, privacy/confidentiality, safer prompting, risk recognition, and professional responsibility.

Sessions 1 and 4 examined participants' reasoning about whether and how to use AI-generated student-support strategies. Coding focused on whether participants identified the suggestions as generic or insufficiently individualized and whether they described concrete checking or contextualization steps before use. Session 2 focused on privacy and ethical boundaries, including whether participants identified identifiable-information concerns and rewrote the prompt in a safer way. Session 3 focused on responsible AI design, including whether participants provided a rule to prevent mistakes, identified risks, and assigned final responsibility to the human professional. Session 4 also included an explicit item asking who was responsible for the final decision.

A detailed coding rubric was developed for all scenario-based indicators reported in the study. For each indicator, the rubric specified the response unit, operational definition, criteria for assigning scores of 1 and 0, treatment of ambiguous or missing responses, and illustrative examples. All indicators were coded dichotomously. The Supplementary Material D identifies the 13 indicators included in the exploratory scenario-based reasoning index and distinguishes them from indicators used only for descriptive reporting. The 13 index indicators were equally weighted because the index was intended as a descriptive count of demonstrated reasoning features rather than a validated performance scale.

The index summarized reasoning features demonstrated across four training sessions and was not designed as a direct measure of pre–post performance change or independently validated professional competence.

3.3.6 Data analysis

Analyses proceeded in four steps. First, participant background characteristics were summarized using frequencies and percentages. Because the AI-use area item allowed multiple selections, percentages for that item were calculated based on the full sample and were not expected to sum to 100%.

Second, descriptive pre- to post-training changes were examined for the three calibrated AI readiness domains—verification orientation, ethical boundary clarity, and accountability ownership—and for the overall nine-item calibrated AI readiness composite. For each score, we reported the pre-training mean and standard deviation, post-training mean and standard deviation, mean change, median change, and the number and percentage of participants whose scores increased, remained the same, or decreased. Given the small sample and single-group design, the analyses emphasized descriptive change patterns rather than causal inference.

Third, scenario-based responses were coded using the finalized rubric and summarized with frequencies and percentages. Two coders independently coded all 25 participants' scenario responses using the 13-indicator rubric. Across indicators with sufficient variation, percentage agreement ranged from 88.0% to 100.0%, and Cohen's κ ranged from.68 to 1.00. Cohen's κ was not estimable for identifying a privacy or confidentiality concern, removing a direct identifier, defining a school-based AI assistant task, assigning final responsibility to a human professional in Session 3, and assigning final decision-making responsibility to a human actor in Session 4 because both coders assigned the same code to all participants. Percentage agreement was 100.0% for these indicators. Agreement for the total scenario-based reasoning score was excellent, ICC(2,1) = .96, 95% CI [.91, .98]. Disagreements were resolved through discussion after reliability was calculated and before the final frequencies and index scores were produced.

Fourth, mixed-methods integration compared self-reported change with scenario-based reasoning. For each participant, an overall self-reported change score was calculated by subtracting the pre-training calibrated AI readiness composite from the post-training composite. The scenario-based reasoning index was calculated by summing 13 equally weighted binary indicators across the four sessions, producing a possible score from 0 to 13. The index was analyzed as a continuous score ranging from 0 to 13, with higher scores indicating more complete demonstration of responsible AI reasoning. No categorical cutoffs were applied. Spearman's rank-order correlation was used to examine the association between self-reported change and scenario-based reasoning.

4 Results

4.1 Phase 1 SEM results

4.1.1 Descriptive statistics and correlations

Table 3 presents descriptive statistics and bivariate correlations for the corrected Phase 1 variables. Participants reported high AI-related epistemic vigilance capacity ($M = 4.48$, $SD = 0.67$). Among the other predictors, autonomous regulatory motivation had a mean of 3.54 ($SD = 1.19$), whereas governance clarity was comparatively lower ($M = 3.30$, $SD = 1.22$). The calibrated AI readiness domain means were 4.19 ($SD = 0.82$) for verification orientation, 4.09 ($SD = 0.95$) for ethical boundary clarity, and 4.47 ($SD = 0.76$) for accountability ownership.

TABLE 3 Means, standard deviations, and correlations among phase 1 study variables.

Variable	*N*	M	SD	1	2	3	4	5	6
1. AI-related epistemic vigilance capacity	425	4.48	0.67	–					
2. Autonomous regulatory motivation	420	3.54	1.19	.17***	–				
3. Governance clarity	425	3.30	1.22	.18***	.16**	–			
4. Verification orientation	425	4.19	0.82	.52***	.15**	.30***	–		
5. Ethical boundary clarity	425	4.09	0.95	.44***	.17***	.36***	.56***	–	
6. Accountability ownership	425	4.47	0.76	.57***	.23***	.27***	.61***	.58***	–

$N = 425$. Correlations are Pearson correlation coefficients.
**$p < .01$.
***$p < .001$.

AI-related epistemic vigilance capacity was positively correlated with verification orientation ($r = .52$, $p < .001$), ethical boundary clarity ($r = .44$, $p < .001$), and accountability ownership ($r = .57$, $p < .001$). Governance clarity was also positively correlated with the three readiness domains ($r = .27$–$.36$, all $p < .001$). Autonomous regulatory motivation showed smaller positive correlations with verification orientation, ethical boundary clarity, and accountability ownership ($r = .15$–$.23$, all $p < .01$). Correlations among the readiness domains ranged from.56 to.61, whereas correlations among the three predictors ranged from.16 to.18.

4.1.2 Model comparison and structural results

The proposed second-order measurement model demonstrated acceptable overall fit (see Table 4), scaled $\chi^2(126) = 275.51$, $p < .001$, robust CFI = .943, robust TLI = .931, robust RMSEA = .060, 90% CI [.050, .070], and SRMR = .049. The correlated six-factor model fit similarly. In contrast, the model combining epistemic vigilance and verification orientation showed poorer fit, as did the model representing the nine readiness items as a single factor. These comparisons supported the proposed multidimensional structure and the treatment of epistemic vigilance and verification orientation as related but distinguishable constructs.

All standardized item loadings (see Table 5) were statistically significant and ranged from.603 to.861. The second-order loadings were.921 for verification orientation, .819 for ethical boundary clarity, and.884 for accountability ownership. Composite reliability ranged from.694 to.856, and AVE ranged from.432 to.666. Verification orientation had comparatively weaker reliability and convergent validity evidence, with composite reliability of.694 and AVE of.432. The remaining constructs had composite reliability values from.778 to.856 and AVE values from.537 to.666.

All HTMT values were below.85, including the epistemic vigilance–verification value of.715. However, the Fornell–Larcker criterion was not fully satisfied for verification orientation. Its square root of AVE (.657) was lower than its latent correlations with ethical boundary clarity (.767), accountability ownership (.808), and epistemic vigilance (.703). Thus, the model comparisons and HTMT results supported construct separation overall, but the convergent and discriminant validity evidence for verification orientation remained mixed (details see Supplementary Material C).

In the primary structural model (see Table 6), AI-related epistemic vigilance capacity was the strongest positive predictor of calibrated AI readiness ($\beta = .675$, $p < .001$), followed by governance clarity ($\beta = .270$, $p < .001$). Autonomous regulatory motivation was not a statistically significant predictor ($\beta = .076$, $p = .156$). Together, the three predictors explained 64.2% of the variance in calibrated AI readiness.

A WLSMV sensitivity analysis was conducted to examine whether treating the five-point indicators as ordered categorical variables changed the findings. The model converged, passed the post-estimation check, and produced no estimation warnings. The scaled fit indices were CFI = .961, TLI = .952, RMSEA = .067, 90% CI [.059, .075], and SRMR = .064. The categorical robust corrections were less favorable, with CFI = .894, TLI = .872, and RMSEA = .102, 90% CI [.090, .114], indicating mixed evidence regarding absolute model fit under the ordinal specification.

The structural conclusions nevertheless remained consistent with the primary MLR results. Epistemic vigilance remained positively associated with calibrated AI readiness ($\beta = .740$, $p < .001$), governance clarity remained positively associated with readiness ($\beta = .277$, $p < .001$), and autonomous regulatory motivation remained nonsignificant ($\beta = .050$, $p = .269$). Standardized item loadings ranged from.730 to.882, and the second-order loadings ranged from.840 to.920. Thus, the directions and statistical-significance conclusions of the structural paths were robust to treating the indicators as ordinal.

4.1.3 Exploratory moderation analysis

To address the exploratory research question (RQ1), the interaction between AI-related epistemic vigilance capacity and autonomous regulatory motivation was examined using observed composite scores. The interaction was not statistically significant (see Table 7), B = −0.020, SE = 0.044, z = −0.441, $p = .659$, 95% CI [−0.106, 0.067], $\beta = -.024$. Adding the interaction increased the explained variance from $R^2 = .431$ to $R^2 = .432$, representing a change of less than.001. Thus, the exploratory moderation analysis did not identify a statistically significant interaction between epistemic vigilance capacity and autonomous regulatory motivation in predicting calibrated AI readiness. Given the use of observed composite scores, this finding should not be interpreted as definitive evidence that moderation is absent.

TABLE 4 Model fit indices for alternative measurement models.

Model	Scaled χ^2	df	Robust CFI	Robust TLI	Robust RMSEA	SRMR	AIC	BIC
Proposed second-order model	275.51	126	.943	.931	.060	.049	19,689.92	19,945.21
Correlated six-factor model	266.25	120	.944	.929	.060	.046	19,690.24	19,969.83
EVC–verification combined model	335.22	125	.919	.901	.072	.054	19,774.60	20,033.93
Single readiness-factor model	383.38	129	.900	.882	.078	.055	19,831.96	20,075.08

TABLE 5 Standardized loadings and measurement quality.

Construct	Standardized loadings	α	Composite reliability	AVE
Calibrated AI readiness	Verification = .921; Ethics = .819; Accountability = .884	–	–	–
Verification orientation	.603, .700, .662	.679	.694	.432
Ethical boundary clarity	.740, .827, .723	.805	.797	.574
Accountability ownership	.641, .836, .795	.791	.821	.604
AI-related epistemic vigilance capacity	.717, .750, .732	.774	.778	.537
Governance clarity	.817, .773, .809	.840	.841	.638
Autonomous regulatory motivation	.861, .807, .779	.854	.856	.666

TABLE 6 Structural paths predicting calibrated AI readiness.

Predictor	B	SE	β	z	*p*
AI-related epistemic vigilance capacity	1.128	.214	.675	5.274	<.001
Governance clarity	.452	.107	.270	4.227	<.001
Autonomous regulatory motivation	.128	.090	.076	1.420	.156

$N = 425$. The model was estimated using MLR with FIML. The predictors explained 64.2% of the variance in calibrated AI readiness.

4.2 Phase 2 pilot training results

4.2.1 Pre- and post-training descriptive changes

Table 8 presents descriptive pre- and post-training changes in participants' self-reported calibrated AI readiness. Participants reported higher post-training scores across verification orientation, ethical boundary clarity, and accountability ownership. The overall nine-item calibrated AI readiness composite increased descriptively from M = 4.20 (SD = 0.52) to M = 4.84 (SD = 0.18). Twenty-one participants (84.0%) showed an increase in the overall composite, one (4.0%) remained unchanged, and three (12.0%) showed a decrease. However, the relatively high baseline scores and post-training scores near the upper end of the five-point scale suggest potential ceiling effects. Given the small sample and absence of a comparison group, these findings should be interpreted as preliminary descriptive changes rather than evidence of training effectiveness.

4.2.2 Scenario-based reasoning

Scenario-based reasoning indicators are presented in Table 9. Across the four embedded training tasks, participants showed applied reasoning related to verification, privacy protection, safer prompting, risk recognition, contextualization, and accountability ownership.

Sessions 1 and 4 provided the most directly comparable evidence of participants' reasoning about AI-generated student-support strategies. In both sessions, 13 participants indicated that they would use the AI-generated suggestions or use them with caution. However, the proportion identifying the suggestions as generic, vague, or insufficiently individualized increased from 72.0% in Session 1 to 80.0% in Session 4. Similarly, the proportion describing a concrete checking or contextualization step increased from 52.0% in Session 1 to 68.0% in Session 4. This pattern suggests that the training was not associated with a simple rejection of AI use, but rather with more cautious, contextualized, and verification-oriented reasoning.

Session 2 provided the clearest evidence of privacy-oriented reasoning. All participants identified a privacy, confidentiality, or identifiable-information concern, and all removed the student's name when rewriting the prompt. However, fewer participants removed diagnostic or disability-related language, suggesting that participants were more consistent in recognizing direct identifiers than in recognizing sensitive educational or diagnostic information.

In Session 3, all participants designed a school-based AI assistant task and assigned final responsibility to the human professional. Most participants also provided a substantive rule or constraint to reduce mistakes. The most identified risks involved hallucination, inaccuracy, poor alignment, over-reliance, automation bias, and lack of contextual fit. Overall, the scenario responses suggest that participants generally understood AI as a support tool requiring human oversight, verification, and contextual judgment rather than as a source of final professional decisions.

4.2.3 Mixed-methods integration

Mixed-methods integration compared each participant's continuous self-reported change score with the continuous scenario-based reasoning index (see Table 10). Observed scenario-based reasoning scores ranged from 8 to 13.

TABLE 7 Exploratory observed-score moderation analysis predicting calibrated AI readiness.

Predictor	B	Robust SE	β	z	*p*	95% CI
Epistemic vigilance capacity	.562	.047	.524	11.935	<.001	[.470, .655]
Autonomous regulatory motivation	.046	.027	.076	1.677	.094	[−.008, .100]
Governance clarity	.156	.026	.267	5.987	<.001	[.105, .207]
Epistemic vigilance × autonomous regulatory motivation	−.020	.044	−.024	−0.441	.659	[−.106, .067]

Predictors were mean-centered before creating the interaction term. The model was estimated using MLR with FIML. $R^2 = .432$; the interaction explained less than.001 additional variance.

TABLE 8 Pre- and post-training descriptive changes in training-aligned responsible AI self-reports.

Domain	*k*	Pre *M* (*SD*)	Post *M* (*SD*)	*M* change	Median change	Increased *n* (%)	Same *n* (%)	Decreased *n* (%)
Verification orientation	3	4.03 (0.85)	4.76 (0.34)	0.73	0.67	18 (72.0)	3 (12.0)	4 (16.0)
Ethical boundary clarity	3	4.17 (0.98)	4.87 (0.27)	0.69	0.33	16 (64.0)	6 (24.0)	3 (12.0)
Accountability ownership	3	4.39 (0.64)	4.89 (0.19)	0.51	0.33	16 (64.0)	6 (24.0)	3 (12.0)
Overall calibrated AI readiness composite	9	4.20 (0.52)	4.84 (0.18)	0.64	0.67	21 (84.0)	1 (4.0)	3 (12.0)

$N = 25$. Items were coded from 1 = disagree to 5 = agree. Each domain score was calculated by averaging its three corresponding items. The overall calibrated AI readiness composite was calculated by averaging all nine items across verification orientation, ethical boundary clarity, and accountability ownership. Change scores were calculated as post-training minus pre-training scores. Increased, same, and decreased classifications were based on participants' unrounded scores.

TABLE 9 Scenario-based reasoning across training sessions.

Session	Scenario focus	Descriptive coded pattern	*n*	%
Session 1	Verification of AI-generated student-support strategies	Willing to use suggestions or use with caution	13	52.0
		Identified suggestions as generic, vague, or insufficiently individualized	18	72.0
		Described a concrete step before use, such as reviewing data, consulting others, or checking fit	13	52.0
Session 2	Ethical use of AI with student information	Judged the prompt inappropriate as written	18	72.0
		Identified privacy, confidentiality, or identifiable-information concern	25	100.0
		Removed the student's name from the rewritten prompt	25	100.0
		Removed diagnostic/disability-related language from the rewritten prompt	15	60.0
		Described concrete checking or contextualization before using AI suggestions	17	68.0
Session 3	Designing a responsible AI assistant	Designed a school-based AI assistant task	25	100.0
		Focused the assistant on lesson planning or instructional design	15	60.0
		Provided a substantive rule or constraint to prevent mistakes	20	80.0
		Identified hallucination, inaccuracy, or poor alignment as a risk	12	48.0
		Identified over-reliance or automation bias as a risk	10	40.0
		Identified lack of contextual fit as a risk	10	40.0
		Assigned final responsibility to the human professional	25	100.0
Session 4	Post-training scenario evaluating AI-generated strategies	Willing to use suggestions or use with caution	13	52.0
		Identified suggestions as generic, vague, or insufficiently individualized	20	80.0
		Mentioned evidence, research, sources, policy, guidelines, or credibility	12	48.0
		Described concrete checking or contextualization before using AI suggestions	17	68.0
		Assigned final responsibility to a human professional, educator, user, or school team	25	100.0

$N = 25$. Percentages are based on the full sample. Rows in this table summarize selected descriptive coded patterns across the four training sessions and are not mutually exclusive. This table is broader than the 13-indicator scenario-based reasoning index used for mixed-methods integration. Descriptive patterns such as willingness to use AI suggestions and task focus were reported for interpretation but were not included in the continuous index. Some related codes, such as different types of AI-use risks in Session 3, were collapsed into a broader risk-recognition indicator for scoring. The continuous scenario-based reasoning index ranged from 0 to 13, with higher scores indicating stronger scenario-based reasoning. The consolidation and scoring procedures are provided in Supplementary Material A.

TABLE 10 Integration of self-reported change and scenario-based reasoning.

Evidence source	Continuous summary	Integrated interpretation
Self-reported calibrated AI readiness	Overall mean increased descriptively from 4.2 before training to 4.84 after training; 21 of 25 participants reported higher post-training scores.	Self-reported scores indicated descriptive change but showed substantial ceiling effects.
Scenario-based reasoning	The index had a possible range of 0–13 and an observed range of 8–13.	Scenario responses provided applied evidence of verification, privacy protection, contextualization, and accountability ownership.
Cross-method relationship	Spearman's $\rho = -.02$, $p = .906$.	Self-reported change and demonstrated reasoning did not converge in this pilot sample.

$N = 25$. Self-reported change was calculated by subtracting the pre-training calibrated AI readiness composite from the post-training composite. The scenario-based reasoning index summed 13 equally weighted binary indicators across Sessions 1–4, with a possible range of 0–13. The index was analyzed continuously; no validated performance thresholds were applied.

No validated performance thresholds were applied. The association between self-reported change and scenario-based reasoning was essentially zero, Spearman's $\rho = -.02$, $p = .906$, indicating no evidence of convergence between the two measures in this pilot sample. Because of the small sample and restricted variation in self-report scores, this result should be interpreted as exploratory.

4.3 Summary of results

The Phase 1 results provided partial support for the hypothesized model. H1 was supported: AI-related epistemic vigilance capacity was a significant positive predictor of calibrated AI readiness. H3 was also supported: governance clarity significantly predicted calibrated AI readiness after accounting for AI-related epistemic vigilance capacity and autonomous regulatory motivation. In contrast, H2 was not supported because autonomous regulatory motivation was not a significant direct predictor in the structural models. Regarding the exploratory research question (RQ1), the observed-score moderation analysis did not identify a statistically significant interaction between AI-related epistemic vigilance capacity and autonomous regulatory motivation.

Given the methodological limitations of using observed composite scores, this result should be interpreted as an absence of evidence for moderation in the present analysis rather than evidence that autonomous regulatory motivation does not have a moderating role. Therefore, these findings suggest that calibrated AI readiness was more strongly associated with professionals' perceived capacity to critically evaluate AI outputs and with the clarity of governance structures than with autonomous regulatory motivation alone.

5 Discussion

This study examined a conceptual model of calibrated AI readiness among K–12 school-based professionals. In Phase 1, AI-related epistemic vigilance capacity and governance clarity were positively associated with self-reported calibrated AI readiness, whereas autonomous regulatory motivation was not a significant direct predictor. The exploratory moderation analysis also did not identify a statistically significant interaction. Phase 2 showed descriptive increases in self-reported readiness following a one-month pilot training program, while scenario-based assessments provided separate evidence of applied responsible AI reasoning. However, self-reported changes and scenario-based reasoning were not significantly associated. Together, these findings highlight the relevance of individual evaluative capacity and organizational guidance, as well as the importance of assessing perceived readiness and applied reasoning separately.

5.1 Epistemic vigilance and governance clarity associated with calibrated AI readiness

One of the central findings of this study is that AI-related epistemic vigilance capacity showed the strongest positive association with calibrated AI readiness ($\beta = .675$, $p < .001$). This finding is consistent with research on calibrated reliance in human–AI collaboration, which emphasizes that appropriate AI use depends not simply on trusting or adopting AI recommendations, but on evaluating their quality and determining when reliance is warranted (Okamura and Yamada, 2020; Schemmer et al., 2023; Schoeffer et al., 2023). Similarly, Dogru and Krämer (2025) examined the roles of expertise, trust, and self-confidence in appropriate reliance on AI-based decision support. Together, these studies provide a relevant basis for interpreting epistemic vigilance as an evaluative capacity associated with responsible AI engagement. However, whereas previous calibration research has primarily examined reliance in specific decision tasks, the present study focused on school professionals' perceived readiness to verify AI-generated information, recognize ethical boundaries, and retain accountability across professional situations. This distinction extends the discussion of calibrated reliance into K–12 school-based practice while acknowledging that perceived readiness is not equivalent to demonstrated calibration behavior.

The positive association between epistemic vigilance and calibrated AI readiness is also consistent with Sperber et al.'s (2010) theoretical conceptualization of epistemic vigilance as the evaluation of communicated information before accepting it as credible. In the context of generative AI, the present findings suggest that professionals who perceive themselves as more capable of evaluating AI-generated information also tend to report stronger orientations toward verification, ethical boundary recognition, and accountability. Nevertheless, this

relationship requires cautious interpretation because epistemic vigilance capacity and verification orientation share conceptual content. Although the alternative measurement models and HTMT results supported treating these constructs separately, verification orientation demonstrated relatively weaker reliability and convergent validity, and the Fornell–Larcker criterion was not fully satisfied. Therefore, the strong observed association should not be interpreted as conclusive evidence that epistemic vigilance represents a fully independent determinant of calibrated AI readiness. Further measurement research is needed to clarify the boundaries between evaluative capacity and self-reported readiness.

Governance clarity was also positively associated with calibrated AI readiness ($\beta = .270$, $p < .001$), after accounting for epistemic vigilance capacity and autonomous regulatory motivation. This finding aligns with responsible AI governance frameworks emphasizing transparency, privacy protection, human oversight, accountability, and clear organizational expectations (National Institute of Standards and Technology, 2023; UNESCO, 2024). Similarly, Farmer et al. (2025) discussed opportunities, challenges, and ethical considerations associated with AI use in psychological practice, while Papagiannidis et al. (2022) situated responsible AI implementation within organizational governance processes. These perspectives highlight the importance of organizational structures in guiding responsible AI engagement. The present study contributes to this literature by empirically examining governance clarity as a perceived organizational condition associated with calibrated AI readiness among school-based professionals. Rather than treating governance solely as a set of institutional requirements, the findings suggest that professionals' understanding of organizational expectations may be relevant to their readiness to evaluate and use AI responsibly.

The governance finding also highlights a distinction between the presence of AI policies and professionals' perceptions of their clarity. Formal guidance may exist without being sufficiently accessible, specific, or understandable to support professional decisions. In the present study, governance clarity had a lower mean ($M = 3.30$, $SD = 1.22$) than epistemic vigilance capacity ($M = 4.48$, $SD = 0.67$), suggesting that participants reported comparatively less clarity about organizational AI expectations than confidence in their own evaluative capacity. This descriptive pattern, together with the positive structural association, supports examining organizational guidance alongside individual capabilities. However, because both variables were measured through self-report in a cross-sectional survey, the findings cannot establish whether clearer governance improves readiness or whether professionals with greater readiness perceive organizational guidance more favorably.

Taken together, these findings support a sociotechnical interpretation of calibrated AI readiness in which individual evaluative capacity and perceived organizational guidance are both relevant. This interpretation also connects with Rind's (2026) conceptualization of teachers as cognitive orchestrators who retain responsibility for professional judgment when engaging with AI. Rind and Qureshi (2026) further discussed professional agency in AI-mediated teaching environments. From this perspective, epistemic vigilance may help professionals question and evaluate AI outputs, while governance clarity may provide a more understandable organizational context for determining whether and how those outputs should be used. However, the present study did not directly measure professional agency, actual reliance behavior, or the quality of AI-assisted decisions. These connections should therefore be understood as theoretically informed interpretations rather than empirically established mechanisms.

5.2 Interpreting the non-significant role of autonomous regulatory motivation

Contrary to H2, autonomous regulatory motivation was not significantly associated with calibrated AI readiness after accounting for AI-related epistemic vigilance capacity and governance clarity ($\beta = .076$, $p = .156$). The exploratory moderation analysis (RQ1) also did not identify a statistically significant interaction between autonomous regulatory motivation and epistemic vigilance capacity ($\beta = -.024$, $p = .659$). These findings differ from the positive relationship that might be expected based on Self-Determination Theory, which emphasizes the importance of autonomous motivation for meaningful engagement, self-directed learning, and sustained participation in valued activities (Deci and Ryan, 2000, 2008). However, Self-Determination Theory does not necessarily imply that greater motivation will independently predict every form of professional competence or judgment after relevant capacities and contextual conditions are considered.

The present findings should also be distinguished from research examining motivation and engagement with educational technologies. For example, Jansen in de Wal et al. (2014) examined motivational profiles associated with teachers' engagement in professional learning, while Schumann et al. (2025) investigated faculty motivation to participate in a digital health professions educator program. Although these studies highlight the relevance of motivation to professional learning and participation, their outcomes differ from the present study's focus on calibrated AI readiness. Similarly, Kim (2026) examined school leaders' AI readiness, perceptions, and implementation strategies, providing a related educational technology context. The nonsignificant association observed here therefore should not be interpreted as contradicting evidence that autonomous motivation supports engagement or technology-related professional learning. Rather, it suggests that motivation to engage in responsible AI learning and perceived readiness to exercise calibrated professional judgment may represent distinct outcomes.

A different perspective emerges from research on appropriate reliance in human–AI collaboration. Dogru and Krämer (2025) examined how expertise, trust, and self-confidence relate to appropriate reliance on AI-based decision support systems, while Küper and Krämer (2025) and Schemmer et al. (2023) addressed the limitations of relying on favorable attitudes or trust when evaluating AI-assisted decisions. These studies provide a conceptual basis for distinguishing willingness to use AI from the evaluative processes required to determine whether AI-generated advice should be accepted, modified, or rejected. In the present study, epistemic vigilance capacity showed a substantially stronger standardized association with calibrated AI

readiness than autonomous regulatory motivation. This pattern is consistent with the interpretation that perceived evaluative capacity may be more proximal to self-reported calibrated readiness than general motivation to engage responsibly with AI. Nevertheless, the comparison remains indirect because these studies examined different constructs, populations, and measures of AI reliance.

Several alternative explanations also warrant consideration. Autonomous motivation may influence responsible AI engagement indirectly through professional learning, sustained verification effort, or opportunities to practice critical evaluation—pathways that were not tested in the present model. In addition, the shared self-report format and the conceptual proximity between epistemic vigilance and verification orientation may influence the relative magnitude of the observed associations. The nonsignificant interaction should also be interpreted cautiously because it was estimated using observed composite scores rather than a latent interaction model. Future research could examine longitudinal or mediated pathways linking autonomous motivation, participation in AI-related professional learning, evaluative capacity, and demonstrated calibration behavior. Thus, the present findings do not establish that autonomous motivation is unimportant; instead, they suggest that its role in calibrated AI readiness may be more complex than the direct and moderating relationships examined here.

5.3 Interpreting the pilot training

Phase 2 provided preliminary evidence of descriptive increases in self-reported verification orientation, ethical boundary clarity, and accountability ownership following the one-month pilot training. These patterns are consistent with the emphasis on critical evaluation, ethical awareness, and human oversight in responsible AI education (Ma et al., 2025; UNESCO, 2024). However, unlike studies designed to establish training effectiveness, the present pilot used a small single-group pretest–posttest design without a comparison group. Therefore, the observed changes cannot be attributed causally to the training. High baseline scores and potential ceiling effects further limit interpretation.

An important finding was the lack of a statistically significant association between changes in self-reported readiness and scenario-based reasoning (Spearman's $\rho = -.02$, $p = .906$). This result is relevant to previous human–AI collaboration research emphasizing appropriate reliance in actual decision-making rather than attitudes or confidence alone (Dogru and Krämer, 2025; Schemmer et al., 2023). Although participants generally demonstrated verification, privacy awareness, and human accountability in scenario responses, these reasoning patterns did not consistently correspond with larger self-reported increases. This divergence suggests that perceived readiness and demonstrated reasoning may capture different aspects of responsible AI engagement.

The findings also extend discussions of responsible AI literacy by highlighting the potential value of incorporating applied scenario-based assessments into professional learning evaluation (Ma et al., 2025). However, the scenario-based reasoning index was exploratory, and its relationship with self-reported change was examined in only 25 participants. Future studies should combine self-report measures with validated performance-based assessments and comparison-group designs to better evaluate whether professional learning improves calibrated AI decision-making.

5.4 Implications for K-12 school-based practice and professional development

For K–12 professionals, effective AI training should focus not only on tool use but also on verification, privacy protection, contextual adaptation, critical evaluation of outputs, and accountability.

Professional development should therefore move beyond general warnings about AI errors and provide structured practice in critical evaluation. Participants can check AI-generated information against reliable sources. They can look for assumptions and possible errors. They can explain whether the information should be used, changed, or rejected. They can also describe how their knowledge of the situation influenced the final decision. These approaches reflect argumentation, claim–evidence coordination, and metacognitive scaffolding recommended by Rind et al. (2026). Training should help professionals distinguish routine tasks that can be delegated to AI from decisions that require human judgment. It should also strengthen their ability to critically evaluate and override AI recommendations when appropriate (Rind, 2026; Rind and Qureshi, 2026).

At the organizational level, school districts and professional associations should provide clear guidelines about permissible AI tools, prohibited student data inputs, documentation standards for AI-assisted work, and decisions that must remain under human professional authority. These safeguards help prevent AI-generated recommendations from inappropriately influencing educational services, support arrangements, intervention plans, and legally regulated documentation.

5.5 Limitations

Several limitations should be considered when interpreting the findings. First, Phase 1 employed a cross-sectional correlational design, preventing causal conclusions about the relationships among epistemic vigilance capacity, autonomous regulatory motivation, governance clarity, and calibrated AI readiness. Participants were recruited through publicly available school district directories, and participation was voluntary. Consequently, selection and nonresponse biases may limit the generalizability of the findings, particularly if professionals with greater interest in AI were more likely to participate.

Second, the measures were developed specifically for this study and received preliminary expert feedback but did not undergo formal content-validity assessment or cognitive interviewing. Although confirmatory factor analyses provided initial evidence supporting the proposed structure, verification orientation demonstrated relatively weaker composite reliability and convergent validity, as well as overlapping variance with

epistemic vigilance and other readiness dimensions. Furthermore, the constructs were measured through self-reports, which may be influenced by social desirability and shared-method variance. Accordingly, self-reported calibrated AI readiness should not be interpreted as demonstrated professional competence.

Third, Phase 2 involved only 25 participants and used a single-group pretest–posttest design without a comparison group. The descriptive increases in readiness therefore cannot be attributed causally to the training. High baseline scores, ceiling effects, repeated measurement, and possible response-shift effects may have influenced the observed patterns.

Fourth, although the scenario-based assessment provided evidence beyond self-report and demonstrated generally high interrater agreement, the 13-indicator reasoning index was exploratory and had not undergone independent validation. It combined reasoning features demonstrated across four training sessions rather than providing a direct pre–post measure of performance. The small sample and restricted variation further limited the interpretation of its association with self-reported change.

Finally, professional agency, cognitive-load redistribution, and potential deskilling were discussed as theoretically relevant considerations but were not directly measured. Future research should examine these mechanisms using longitudinal, experimental, and performance-based designs with more diverse professional samples.

6 Future research

Future studies can extend the findings from this study in several areas. First, there could be more large-scale randomized training studies to more robustly examine if the calibration-oriented training could enhance professionals' capacity in AI calibration. Second, longitudinal studies could be helpful to test if governance clarity and epistemic vigilance could predict continuous changes in the daily practices of school professionals. Third, from the measurement perspective, future studies should further examine dimensional structure, discriminant validity, and measurement invariance of calibrated AI readiness with a wide range of samples from different professional roles and organizational contexts. Item-level content-validity indices and cognitive interviews should be conducted to investigate the item relevance, construct coverage, comprehension, and response processes. At last, practically, future research should examine how AI training for professional development in K–12 schools supports or potentially disadvantages some professionals from under-resourced schools.

7 Conclusion

This study explored the individual and organizational factors that are associated with K-12 school professionals' calibrated AI readiness. The quantitative survey results show that high level of AI-related epistemic vigilance capacity and clear governance clarity are strongly associated with high level of calibrated AI readiness. In comparison, autonomous regulatory motivation has limited independent association with calibrated AI readiness with the sample in this study. Even though the pilot training results show that participants show positive descriptive changes in self-reports and scenario responses, given the limited sample size in pilot and lack of control group, the results cannot be interpreted as effective training.

Data availability statement

The raw data supporting the conclusions of this article will be made available by the authors, without undue reservation.

Ethics statement

The studies involving humans were approved by Committee for the Protection of Human Subjects (CPHS), Dartmouth College, Hanover, New Hampshire, USA. The studies were conducted in accordance with the local legislation and institutional requirements. The participants provided their written informed consent to participate in this study.

Author contributions

ZS: Software, Writing – original draft, Formal analysis, Funding acquisition, Resources, Visualization, Project administration, Supervision, Methodology, Writing – review & editing, Data curation, Conceptualization, Investigation, Validation.

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The author(s) declared that this work was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

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The author(s) declared that generative AI was used in the creation of this manuscript. The author used OpenAI ChatGPT (GPT-5.5 Thinking, OpenAI, 2026) for language revision, and manuscript organization. The author reviewed and verified the accuracy of the content, analyses, citations, and references.

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Supplementary material

The Supplementary Material for this article can be found online at: https://www.frontiersin.org/articles/10.3389/feduc.2026.1946896/full#supplementary-material

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